Kevin Valencia

Education

University of Michigan, Ph.D. in Computer Science and Engineering — Sept 2026 – Present

- Advisors: Lauren Gillespie and Jeong Joon (JJ) Park

University of California, Los Angeles, CA, B.S. in Computer Science — Sept 2024 – June 2026

- GPA: 3.93/4.0

De Anza College, Cupertino, CA, A.S. in Computer Science; A.S. in Mathematics; A.S. in Liberal Arts — Sept 2022 – June 2024

- GPA: 4.0/4.0

Research Experience

Graduate Student Researcher, University of Michigan, Ann Arbor, MI — Sept 2026 – Present

- Investigating geospatial foundation models for ecological mapping in the GAIEA Lab, using expert field surveys and hierarchical vegetation classifications to evaluate fine-grained ecological information in learned representations.
- Exploring physics- and observation-grounded generative modeling for weather and climate extremes in the Park Lab, with an emphasis on inference-time physical constraints, sparse observations, and high-resolution downscaling.

Research Assistant, NASA Jet Propulsion Laboratory, Pasadena, CA — April 2025 – May 2026

- Developed geospatial processing and learning pipelines for coral bleaching detection using multispectral PlanetScope SuperDove satellite imagery.
- Evaluated reef-scale coral health classification across geographically distinct regions, including reefs held out from training, to study model generalization.

Research Intern, Brookhaven National Laboratory, Upton, NY — June 2025 – Aug 2025

- Led development of **OmniField**, a conditioned neural field for reconstructing and forecasting continuous scientific fields from sparse, irregular multimodal measurements; published at ICLR 2026.
- Developed **SOLID**, a diffusion-based approach to learning spatiotemporal fields from sparse supervision, supporting uncertainty-calibrated reconstruction and forecasting.

Undergraduate Research Assistant, UCLA Computational Machine Learning Lab — Feb 2025 – Aug 2025

- Studied data-efficient diffusion models for ensemble weather forecasting using ERA5, evaluating training-data selection and sampling strategies.
- Contributed to research on text preservation in text-to-video generation using synthetic text-rich videos; both projects led to ICCV 2025 workshop publications.

Research Intern, Stanford SURGE Program, Stanford University, Stanford, CA — June 2023 – Jan 2024

- Explored multimodal generative modeling for sustainable architectural design, combining sketches and text with structured 3D outputs in NVIDIA Omniverse workflows.

Peer-Reviewed Publications

K. Valencia, T. Balasooriya, X. Luo, S. Yoo, D. Park. "OmniField: Conditioned Neural Fields for Robust Multimodal Spatiotemporal Learning." *International Conference on Learning Representations (ICLR)*, 2026.

K. Valencia, Z. Liu, J. Cui. "Data-Efficient Ensemble Weather Forecasting with Diffusion Models." *ICCV Workshop on Curated Data for Efficient Learning*, 2025.

T. Balasooriya, J. Choi, **K. Valencia**, X. Luo, S. Yoo, D. Park. "Mamba-GINR: A Scalable Framework for Spatiotemporal Representation of fMRI." *NeurIPS Workshop on Data on the Brain & Mind*, 2025.

Z. Liu, **K. Valencia**, J. Cui. "Video Text Preservation with Synthetic Text-Rich Videos." *ICCV Workshop on Curated Data for Efficient Learning*, 2025.

Preprints and Manuscripts

K. Valencia, X. Luo, S. Yoo, D. Park. "Uncertainty-Calibrated Spatiotemporal Field Diffusion with Sparse Supervision." *arXiv preprint*, 2026.

M. Ayad, **K. Valencia**, C. Lee, B. Lütjens, R. Kudela. "Detection of Coral Bleaching using PlanetScope Imagery using the CT Classification Model." *Under review, International Journal of Applied Earth Observation and Geoinformation*.

Fellowships and Honors

2026 DOE Computational Science Graduate Fellow – U.S. Department of Energy

2026 Rackham Graduate Merit Fellow – University of Michigan

2026 GEM Fellow – National GEM Consortium

2025 HSF Scholar – Hispanic Scholarship Fund

2025 Coding it Forward Discovery Program Fellow – Coding it Forward

UCLA CEED Scholarship (2025, 2026) – UCLA Center for Excellence in Engineering and Diversity

De Anza College President's Award – Awarded to two students annually

De Anza Kiwanis Club Scholarship – De Anza College

Leadership and Mentorship

Board Intern, Society of Latinx Engineers and Scientists (SOLES), UCLA 2024 – 2025

- Supported industry events, workshops, and tutoring programs for Latinx students in engineering and computing.

Student Ambassador, De Anza College Men of Color Community 2022 – 2024

- Led mentoring and student-support workshops and represented students in conversations with campus leadership.

Outreach Assistant, De Anza College Office of Outreach 2022 – 2024

- Conducted school outreach, campus tours, and enrollment presentations to expand access to higher education.